دانشگاه صنعتی خواجه نصیرالدین طوسی

Research Project · Digital image processing course

CARD: Correlation Aware Restoration with Diffusion

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Motivation

Most image restoration methods assume image noise is independent and identically distributed Gaussian noise.

Real camera sensors, especially rolling-shutter CMOS sensors, can produce spatially correlated noise:

- neighboring pixels are statistically coupled;
- row-wise readout can create banding artifacts;
- restoration methods that ignore this structure can leave artifacts or over-smooth details.

Original CARD Paper

The paper proposes **CARD: Correlation Aware Restoration with Diffusion**.

Its main idea is:

$$y = Hx_0 + n, \qquad n \sim \mathcal{N}(0, \sigma_y^2 \Sigma)$$

CARD whitens the measurement:

$$\tilde{y} = \Sigma^{-1/2} y, \qquad \tilde{H} = \Sigma^{-1/2} H$$

After whitening:

$$\tilde{n} \sim \mathcal{N}(0, \sigma_y^2 I)$$

This lets DDRM-style diffusion restoration operate under its usual i.i.d. noise assumption.

What This Project Implements

This project implements a compact, CPU-friendly version of the CARD idea.

- Generate a synthetic test image.
- Build rolling-shutter-like correlated Gaussian noise.
- Construct a patch-local covariance matrix.
- Use covariance coloring to add correlated noise.
- Use covariance eigenbasis / whitening logic for restoration.
- Compare against a Gaussian denoising baseline.

The implementation package is named `card`.

Pipeline

1. Create a clean image.
2. Build covariance matrix Σ.
3. Generate correlated noise using a coloring matrix.
4. Add noise to the image.
5. Restore using:
   - Gaussian baseline;
   - CARD covariance-aware Wiener restoration.
6. Report PSNR and SSIM.
7. Save visual output to `results/card_demo.png`.

Covariance Model

The project models local spatial correlation inside each patch.

$$\Sigma_{ij} = \rho^{|r_i - r_j|}(0.45\rho)^{|c_i - c_j|} + \epsilon \boldsymbol{I}$$

- r_i, c_i: row and column coordinates of pixel i in a patch.
- ρ: correlation strength.
- $\epsilon \boldsymbol{I}$: small regularization for numerical stability.

This creates stronger row-wise correlation, similar to rolling-shutter artifacts.

Noise Coloring and Whitening

The covariance matrix is decomposed with eigenvalue decomposition:

$$\Sigma = Q \Lambda Q^T$$

Coloring matrix:

$$C = Q \Lambda^{1/2} Q^T$$

Whitening matrix:

$$W = Q \Lambda^{-1/2} Q^T$$

- Coloring transforms i.i.d. Gaussian noise into correlated noise.
- Whitening transforms correlated noise toward an i.i.d. form.

Main Run Command

The project is reproducible with `uv`.

```
uv sync
uv run python main.py
```

Verification:

```
uv run pytest
uv run ruff check .
uv run basedpyright
```

Implementation Structure

File	Responsibility
`card/noise.py`	covariance, coloring, whitening, noise generation
`card/restoration.py`	Gaussian baseline and CARD restoration
`card/metrics.py`	PSNR and SSIM metrics
`card/images.py`	synthetic image and output image grid
`card/pipeline.py`	end-to-end experiment pipeline
`main.py`	command-line entry point

CARD Restoration

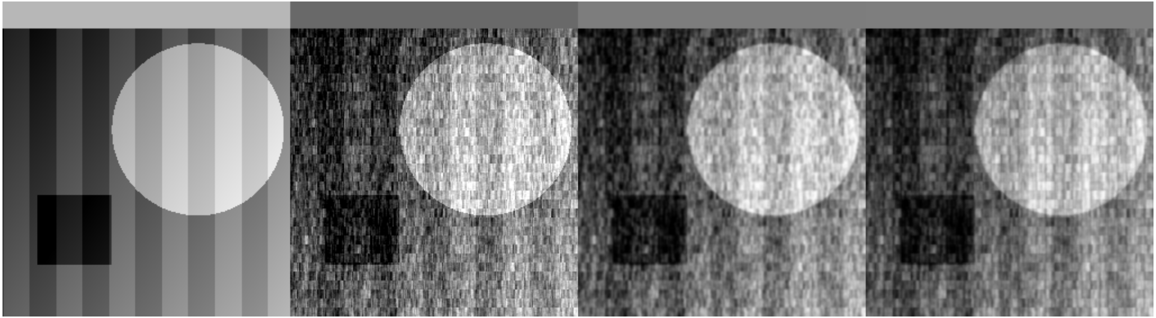
The restoration step uses the covariance eigenbasis.

1. Split the noisy image into patches.
2. Project patch vectors into the covariance eigenbasis.
3. Estimate signal variance and noise variance per component.
4. Apply Wiener-style shrinkage:

$$g_i = \frac{\sigma^2_{\text{signal},i}}{\sigma^2_{\text{signal},i} + \sigma^2_{\text{noise},i}}$$

5. Reconstruct the image from restored patches.
6. Blend the covariance-aware result with a Gaussian image prior.

Result



Left to right: clean image, correlated noisy image, Gaussian baseline, CARD-style restoration.

Measured Output

Image	PSNR	SSIM
Correlated noisy	18.62	0.889
Gaussian baseline	22.07	0.946
CARD-style restore	22.39	0.949

The covariance-aware restoration improves over both the noisy input and the Gaussian baseline in this deterministic demonstration.

Difference from the Original Paper

Component	Original CARD	This Project
Restoration engine	DDRM diffusion sampler	Wiener-style denoising
Learned prior	Pretrained diffusion model	Gaussian image prior
Tasks	Denoising, deblurring, SR	Denoising only
Noise data	Synthetic and real CIN-D	Synthetic only
Covariance	Dark frames or synthetic	Hand-designed synthetic
Evaluation	ImageNet, LSUN, CIN-D	One deterministic demo

Why This Version Is Useful

- It runs quickly on CPU.
- It has no pretrained model dependency.
- The covariance and restoration logic are easy to inspect.
- It demonstrates the core insight of CARD:

  model the noise correlation instead of pretending it is i.i.d.

- It is suitable for a course presentation and reproducible experiment.

Limitations

- It is not a faithful implementation of the full diffusion/DDRM method.
- It does not use real dark-frame calibration.
- It does not evaluate on real camera datasets.
- It does not support deblurring or super-resolution.
- Its SSIM implementation is a simple global metric, not full local-window SSIM.

Conclusion

CARD shows that correlated sensor noise should be handled explicitly.

This project preserves the central image-processing concept:

- estimate or define a covariance matrix;
- use it to color and whiten noise;
- restore in a covariance-aware representation;
- compare with a method that ignores correlation.

The result is a small, reproducible implementation suitable for explaining the method before moving to the full diffusion-based version.